

Pratham Vanangadu Keshava Babu

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SUMMARY

Proven expertise in Python and SQL for data reconciliation and large-scale data analysis, with hands-on experience building data pipelines, deploying LLM-based AI solutions (including RAG and LangChain), and integrating RESTful and SOAP APIs for enterprise systems. Skilled in cross-functional collaboration with clients and internal stakeholders to translate business requirements into scalable data workflows. Experienced in Agile delivery, data quality validation, and production data monitoring. Adept at leveraging AI tools including LLMs and agentic workflows to streamline data processes and accelerate client onboarding outcomes.

EDUCATION

Arizona State University, Tempe, Arizona, USA <i>MS, Systems Engineering (Robotics and Autonomous Systems)</i>	Aug 2023 - May 2025
Vemana Institute of Technology (VTU Affiliated), Bangalore, India <i>B.E, Information and Computer Science</i>	Jul 2018 - Jun 2022

WORK EXPERIENCE

Internpro.ai <i>Robotics Software Engineer</i>	Tempe, United States Aug 2025 – July 2026
- Built real-time distributed data processing pipelines using C++, Python, ROS 2, and sensor fusion technologies, applying object-oriented design principles to support scalable data ingestion and reconciliation workflows, enabling continuous data flow for production systems. - Engineered data validation and anomaly detection algorithms using ROS 2 action servers and asynchronous processing, reducing data processing errors by 95% through optimized fault-tolerant design and automated quality checks. - Developed simulation-to-production deployment workflows using Docker and Gazebo; implemented data transformation pipelines that improved data accuracy by 27% and streamlined client delivery timelines. - Collaborated with cross-functional stakeholders to define data requirements, validate data outputs, and ensure successful adoption of platform data conventions across multiple client environments.	
Ness Digital Engineering <i>Software Engineer</i>	Bengaluru, India Feb 2023 - Aug 2023
- Designed and developed microservices in Java (Spring Boot) and C# using Agile/SCRUM methodologies, improving system scalability and reducing deployment time by 30% through CI/CD pipeline automation (Azure DevOps, GitHub Actions). - Built and integrated RESTful and SOAP APIs handling JSON, XML, and WSDL payloads for enterprise data exchange; implemented secure JWT authentication aligned with identity management protocols, enabling seamless client data onboarding. - Engineered asynchronous, fault-tolerant distributed data pipelines using Docker, Kubernetes, and Azure services (Cosmos DB, Queue Storage, Event Hubs, Data Factory); deployed containerized microservices with real-time data updates via WebSocket and Redis caching. - Performed large-scale data reconciliation and validation using Python and SQL scripts; optimized MongoDB query performance and achieved 85%+ code coverage through comprehensive JUnit and Mockito testing. - Collaborated with business analysts, architects, and clients on requirements gathering, system design, and long-term data strategy; led code reviews using Git and SonarQube, raising code quality standards and reducing production defects.	
Istop.ai <i>Machine Learning Intern</i>	India Aug 2021 - Oct 2021
- Developed and deployed RESTful APIs with Python (FastAPI, Flask) to serve ML models in production; built data-processing pipelines using Docker and Azure SQL, and integrated inference services into client-facing web applications. - Debugged API services, tuned endpoint performance, and performed integration testing with Postman and Docker to achieve reliable, low-latency data delivery for client requests.	

PROJECTS

Swin-B Transformer for Chest X-ray Classification
- Trained a Swin-B transformer (88M params, shifted-window attention) on CheXpert v1.0 (224K radiographs, 65K patients) for multi-label classification across 14 pathologies - Handled label uncertainty (positive/uncertain/negative/not-mentioned) using BCEWithLogitsLoss, AdamW, and ReduceLROnPlateau scheduling over 25 epochs. Applied transfer learning from pre-trained weights, adapting a general-purpose vision backbone to a domain with sparse and noisy ground-truth labels
BART Summarizer with LoRA Fine-Tuning
- Fine-tuned BART-base for abstractive summarization using LoRA, reducing trainable parameters to ~1.2% of the full model while cutting training compute significantly - Demonstrated parameter-efficient fine-tuning as a lightweight alternative to full fine-tuning, matching performance with a fraction of the resource cost
<i>Additional Projects, on model training, fine-tuning and dataset curation.</i>

SKILLS

Programming: Python, Java, Spring Boot, J2EE, C#, C++, TypeScript, JavaScript, FastAPI, Flask Data & Analytics: SQL, MongoDB, PostgreSQL, Redis, Azure SQL, Cosmos DB, Oracle, Excel, Data and ETL Pipelines AI / LLM: Hugging Face, PyTorch, LoRA, BART, RAG, Prompt Eng, LangChain, Agentic Workflows, LLM Fine-Tuning Web Services / APIs: REST, SOAP, JSON, XML, JAXB, gRPC patterns, Postman, JWT Authentication, WSDL Cloud & DevOps: Azure, Docker, Kubernetes, Azure DevOps, GitHub Actions, Terraform basics, Maven, CI/CD Development Tools: Git, Jira, Agile/SCRUM, Linux/Unix shell scripting, IntelliJ, VS Code, SonarQube, Data Factory
